

# **Transcriptomic Profiling of TEX10-Depleted Colorectal Cancer Cells: A GEO2R-Based Differential Gene Expression and KEGG Pathway Enrichment Study**

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Dataset: GSE147334 | Platform: RNA-seq (DESeq2) | HCT116 Colorectal Cancer Cells

## ABSTRACT

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Colorectal cancer (CRC) is among the most prevalent and lethal malignancies worldwide, accounting for approximately 1.9 million new cases and 935,000 deaths annually. TEX10 (Testis-Expressed Gene 10) has been identified as a candidate oncogene that promotes CRC cell proliferation through activation of the NF- $\kappa$ B signaling pathway. In this study, we performed a comprehensive bioinformatics analysis of the publicly available GEO RNA-seq dataset GSE147334, comprising HCT116 colorectal cancer cells subjected to TEX10 knockdown (shRNA) versus scrambled negative control (shNC). Using the GEO2R web tool implemented via DESeq2, 18,903 genes were analyzed across 8 samples (3 TEX10 knockdown vs. 5 shNC). Differential expression analysis identified 235 significant differentially expressed genes (DEGs) at adjusted p-value ( $\text{padj}$ )  $\leq 0.05$  and  $|\log_2\text{FC}| \geq 1$ , comprising 103 upregulated and 132 downregulated genes. The most significantly upregulated gene was HRNR (hornerin;  $\log_2\text{FC} = +3.12$ ,  $\text{padj} = 5.48 \times 10^{-17}$ ) and the most significantly downregulated was HBA2 ( $\log_2\text{FC} = -2.10$ ,  $\text{padj} = 9.50 \times 10^{-11}$ ). Seven quality control visualizations (volcano plot, MA plot, UMAP, Venn diagram, box plot, p-value histogram, dispersion estimates) confirmed data quality and group separation. KEGG pathway enrichment via DAVID identified 10 enriched pathways: 8 from upregulated DEGs (top: Systemic lupus erythematosus and Cornified envelope formation, both  $\text{FDR} = 6.39 \times 10^{-4}$ ) and 2 from downregulated DEGs (Growth hormone synthesis; Wnt signaling). These findings provide mechanistic insight into TEX10-driven transcriptional reprogramming in colorectal cancer.

**Keywords:** *TEX10, colorectal cancer, HCT116, GEO2R, DESeq2, differential gene expression, KEGG, DAVID, NF- $\kappa$ B, RNA-seq, GSE147334*

## 1. INTRODUCTION

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Colorectal cancer (CRC) is the third most commonly diagnosed malignancy and the second leading cause of cancer-related death globally. Despite advances in surgical, chemotherapeutic, and immunotherapeutic approaches, the five-year survival rate for metastatic CRC remains below 15%, underscoring the urgent need for a deeper mechanistic understanding of the molecular drivers governing disease initiation, progression, and therapeutic resistance.

The NF- $\kappa$ B (Nuclear Factor kappa-light-chain-enhancer of activated B cells) signaling pathway is one of the most extensively characterized oncogenic cascades in human cancer. Constitutive NF- $\kappa$ B activation in CRC promotes tumor cell survival, proliferation, angiogenesis, invasion, and resistance to apoptosis. The canonical pathway is activated by a broad spectrum of stimuli, converging on nuclear translocation of NF- $\kappa$ B transcription factor complexes that drive pro-tumorigenic gene expression.

TEX10 (Testis-Expressed Gene 10) was originally described in the context of embryonic stem cell pluripotency regulation. More recently, RNAi screening identified TEX10 as a functionally important gene in CRC biology, with evidence that it promotes proliferation of colorectal cancer cells through enhancement of NF- $\kappa$ B transcriptional activity. Knockdown of TEX10 in HCT116 CRC cells suppresses NF- $\kappa$ B target gene expression and reduces cellular proliferation, positioning TEX10 as a potential therapeutic target. However, the comprehensive transcriptomic landscape regulated by TEX10 in CRC cells remains incompletely characterized.

GEO2R is an interactive web-based analytical tool that enables differential gene expression analysis directly on NCBI GEO datasets using the DESeq2 algorithm — the current gold standard for RNA-seq differential expression analysis, incorporating size factor normalization, dispersion estimation, and Wald test-based hypothesis testing with Benjamini-Hochberg multiple testing correction.

In the present study, we performed a systematic bioinformatics analysis of GEO dataset GSE147334, comparing TEX10 knockdown HCT116 CRC cells against scrambled shRNA controls. Our objectives were: (1) to identify significant DEGs using GEO2R/DESeq2; (2) to visualize differential expression and sample quality through seven complementary plots; (3) to perform

KEGG pathway enrichment via DAVID; and (4) to construct publication-quality KEGG visualizations.

## 2. MATERIALS AND METHODS

### 2.1 Dataset Acquisition

The RNA-seq dataset GSE147334, entitled "RNAi Screening Identifies that TEX10 Promotes the Proliferation of Colorectal Cancer Cells by Increasing NF- $\kappa$ B Activation," was retrieved from NCBI GEO (<https://www.ncbi.nlm.nih.gov/geo/>). The dataset comprises 10 RNA-seq samples from HCT116 human colorectal cancer cells under TEX10 shRNA knockdown, p65 shRNA knockdown, and scrambled shNC conditions. All samples were sequenced using Illumina high-throughput RNA-seq technology.

### 2.2 Group Definition

Samples were classified into two groups. The Tumor Group comprised three TEX10 knockdown replicates (GSM4426224, GSM4426225, GSM4426226). The Control Group comprised five shNC replicates from two constructs: shNC1 (GSM4426221, GSM4426222, GSM4426223) and shNC2 (GSM4426227, GSM4426228), as detailed in Table 1.

| Accession  | Sample Name   | Source / Condition             | Cell Line | Group         |
|------------|---------------|--------------------------------|-----------|---------------|
| GSM4426221 | shNC1 rep1    | HCT116_shNC (Negative Control) | HCT116    | Control Group |
| GSM4426222 | shNC1 rep2    | HCT116_shNC (Negative Control) | HCT116    | Control Group |
| GSM4426223 | shNC1 rep3    | HCT116_shNC (Negative Control) | HCT116    | Control Group |
| GSM4426227 | shNC2 rep1    | HCT116_shNC (Negative Control) | HCT116    | Control Group |
| GSM4426228 | shNC2 rep2    | HCT116_shNC (Negative Control) | HCT116    | Control Group |
| GSM4426224 | TEX10_sh rep1 | HCT116_TEX10 Knockdown         | HCT116    | Tumor Group   |
| GSM4426225 | TEX10_sh rep2 | HCT116_TEX10 Knockdown         | HCT116    | Tumor Group   |
| GSM4426226 | TEX10_sh rep3 | HCT116_TEX10 Knockdown         | HCT116    | Tumor Group   |

Table 1. Sample classification for GEO2R differential expression analysis (GSE147334).

### 2.3 Differential Expression Analysis

Differential expression was performed via GEO2R (<https://www.ncbi.nlm.nih.gov/geo/geo2r/>) using DESeq2 (v1.38+). Parameters:

- P-value adjustment: Benjamini-Hochberg (BH) False Discovery Rate correction
- Significance threshold:  $\text{padj} \leq 0.05$  and  $|\log_2\text{FC}| \geq 1$
- Contrast: Tumor Group (TEX10 knockdown) vs. Control Group (shNC)
- Total genes tested: 18,903 | Platform: RNA-seq (Illumina)

## 2.4 Quality Control Visualizations

Seven plots were generated via GEO2R: (1) Volcano plot —  $\log_2\text{FC}$  vs.  $-\log_{10}(\text{padj})$ ; (2) MA (Mean-Difference) plot —  $\log_2\text{FC}$  vs.  $\log_{10}$  mean counts; (3) UMAP — unsupervised sample clustering; (4) Venn diagram — DEG count; (5) Box plot — normalized count distributions; (6) P-value histogram —  $\text{padj}$  distribution; (7) Dispersion estimates — model fit assessment.

## 2.5 KEGG Pathway Enrichment

Upregulated and downregulated DEG lists were submitted separately to DAVID (v2023, <https://david.ncifcrf.gov/>) using Official Gene Symbols (Homo sapiens). EASE threshold = 0.05, Count Threshold = 2. Results were visualized as bar charts and bubble plots in Python (matplotlib v3.8).

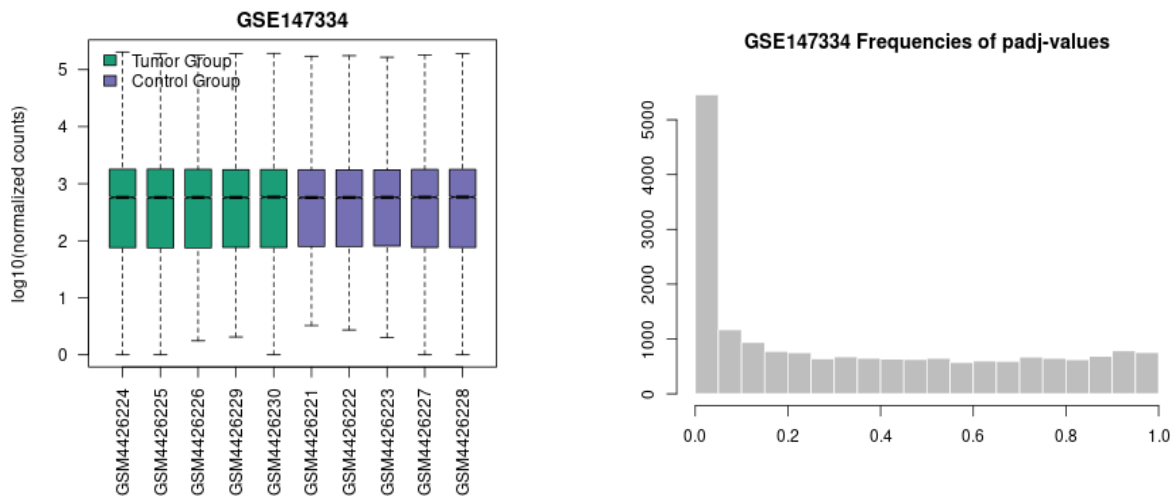
### 3. RESULTS

#### 3.1 Data Quality Assessment

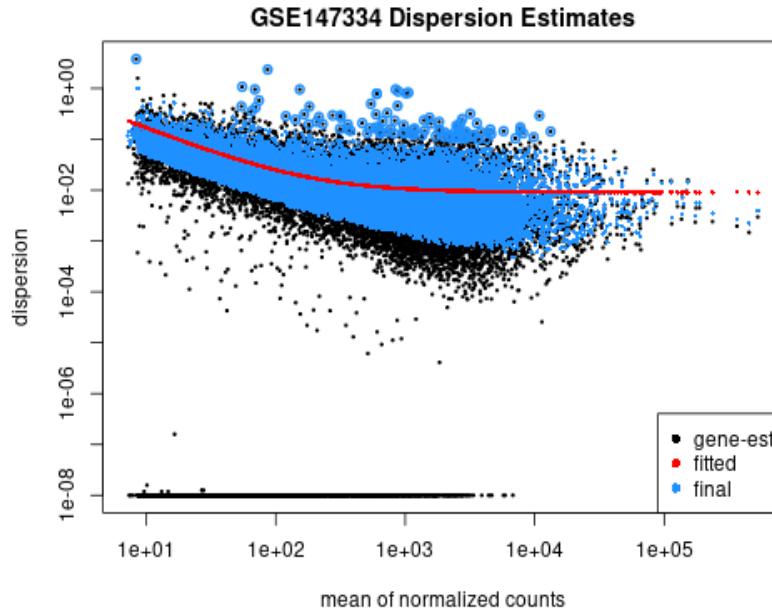
The box plot (Figure 1) shows consistent  $\log_{10}(\text{normalized counts})$  distributions across all 8 samples, with medians centered at 2.5–3.0 for both Tumor Group (green) and Control Group (purple), confirming successful DESeq2 normalization without outliers.

The p-value histogram (Figure 2) shows strong enrichment of low  $p_{\text{adj}}$  values ( $\sim 5,400$  genes near zero) with a relatively uniform distribution at higher values — an ideal pattern confirming genuine differential expression signal and appropriate statistical modeling.

The dispersion estimates plot (Figure 3) demonstrates a well-fitted negative binomial model: gene-level estimates (black) follow the expected decreasing dispersion trend, the fitted curve (red) accurately captures this relationship, and final shrunken estimates (blue) cluster tightly around the trend.



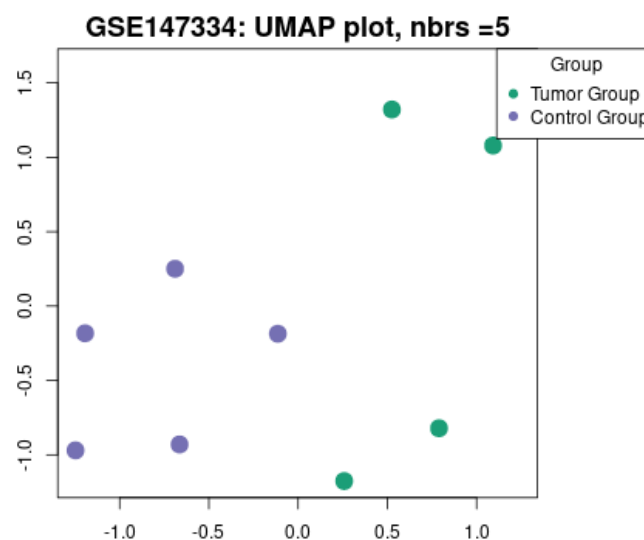
**Figure 1.** Box plot of  $\log_{10}(\text{normalized counts})$  per sample. Green = Tumor Group; Purple = Control Group. Uniform distributions confirm successful normalization. **Figure 2.** Frequency histogram of  $p_{\text{adj}}$  values for all 18,903 tested genes. High bar near zero confirms genuine differential expression signal.



**Figure 3.** Dispersion estimates plot (GSE147334). Black = gene-level estimates; Red curve = fitted trend; Blue = final shrunk estimates. Well-fitted trend confirms appropriate negative binomial model for 18,903 genes.

### 3.2 Sample Clustering — UMAP

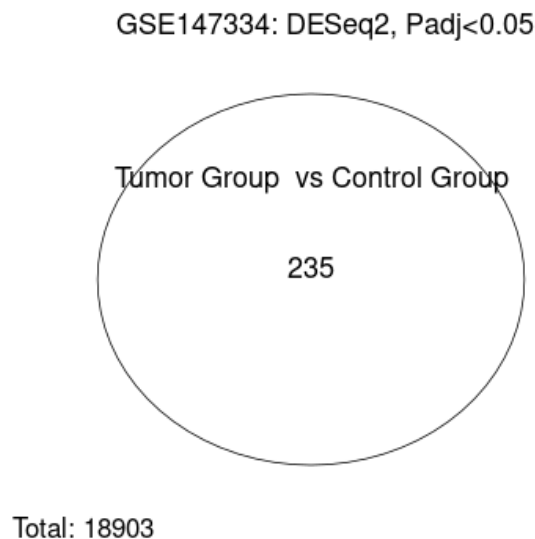
UMAP dimensionality reduction (Figure 4, nbrs = 5) revealed clear spatial separation between the Tumor Group (green, n = 3) and Control Group (purple, n = 5), confirming that TEX10 depletion induces a distinct and reproducible transcriptomic state. Control samples form sub-clusters corresponding to shNC1 and shNC2 constructs.



**Figure 4.** UMAP plot (GSE147334, nbrs = 5). Green = Tumor Group (TEX10 knockdown,  $n = 3$ ); Purple = Control Group (shNC,  $n = 5$ ). Clear separation between groups confirms distinct transcriptomic states induced by TEX10 depletion.

### 3.3 DEG Overview — Venn Diagram

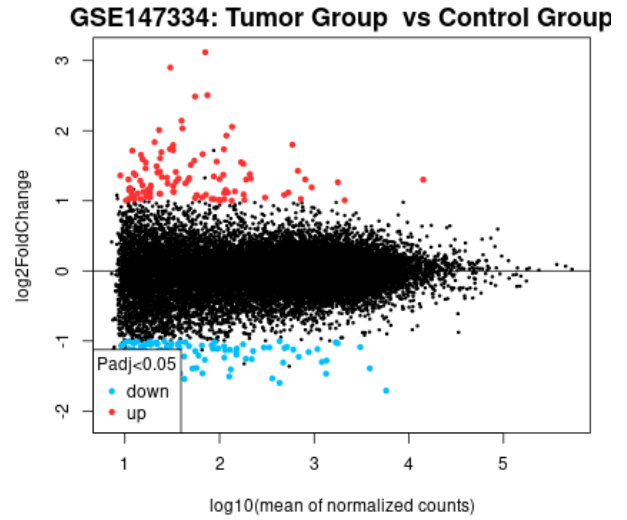
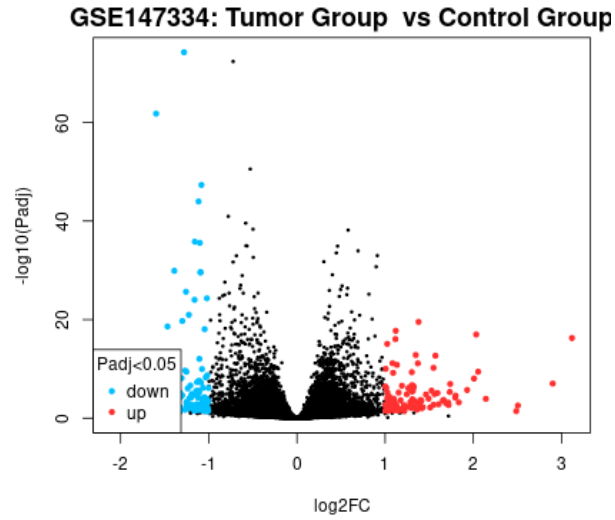
The DESeq2 Venn diagram (Figure 5) shows that of 18,903 total genes tested, 235 DEGs met the  $\text{padj} < 0.05$  threshold in the Tumor Group vs. Control Group contrast, representing 1.24% of all analyzed genes.



**Figure 5.** Venn diagram (DESeq2, GSE147334,  $\text{padj} < 0.05$ ). The single-circle diagram shows 235 significant DEGs identified in the Tumor Group vs. Control Group comparison from 18,903 total tested genes.

### 3.4 Volcano Plot and MA Plot

The volcano plot (Figure 6) displays  $-\log_{10}(\text{padj})$  versus  $\log_2\text{FC}$  for all 18,903 genes. Upregulated DEGs (red;  $\log_2\text{FC} \geq 1$ ) appear in the upper-right quadrant; downregulated DEGs (cyan;  $\log_2\text{FC} \leq -1$ ) in the upper-left. The  $\log_2\text{FC}$  range spans  $-2.10$  to  $+3.12$  for significant genes. The MA plot (Figure 7) confirms DEGs span the full range of mean expression levels without systematic bias toward low- or high-abundance transcripts.



**Figure 6.** Volcano plot (GSE147334: Tumor Group vs. Control Group). X-axis:  $\log_2\text{FC}$ ; Y-axis:  $-\log_{10}(\text{padj})$ . Red = upregulated DEGs; Cyan = downregulated DEGs; Black = not significant.

**Figure 7.** MA plot (Mean-Difference plot, GSE147334). X-axis:  $\log_{10}(\text{mean normalized counts})$ ; Y-axis:  $\log_2\text{FC}$ . Red = upregulated; Cyan = downregulated at  $\text{padj} < 0.05$ .

### 3.5 Top Differentially Expressed Genes

Table 2 summarizes DEG counts. Table 3 presents the top 5 upregulated and top 5 downregulated genes. HRNR (hornerin;  $\log_2FC = +3.12$ ,  $padj = 5.48 \times 10^{-17}$ ) was most upregulated, followed by FLG (+2.90), GJA1 (+2.51), CEND1 (+2.49), and SCN2A (+2.14). HBA2 showed the greatest downregulation ( $\log_2FC = -2.10$ ,  $padj = 9.50 \times 10^{-11}$ ), followed by HBA1 (-1.82), LINC00887 (-1.80), LCNL1 (-1.78), and NDNF (-1.78). The transcription factor EGR1 ( $baseMean = 5,763$ ;  $\log_2FC = -1.71$ ,  $padj = 1.14 \times 10^{-12}$ ) was among the most significantly altered high-expression genes.

| Category                                                    | Count          | Percentage (%) |
|-------------------------------------------------------------|----------------|----------------|
| Total genes tested                                          | 18,903         | 100.0          |
| Significant DEGs ( $padj \leq 0.05$ , $ \log_2FC  \geq 1$ ) | 235            | 1.24           |
| Upregulated ( $\log_2FC \geq +1$ )                          | 103            | 43.8           |
| Downregulated ( $\log_2FC \leq -1$ )                        | 132            | 56.2           |
| $\log_2FC$ range (upregulated)                              | +1.00 to +3.12 | —              |
| $\log_2FC$ range (downregulated)                            | -2.10 to -1.00 | —              |

Table 2. Summary of GEO2R/DESeq2 results for GSE147334 (Tumor Group vs. Control Group).

| Symbol    | Dir. | $\log_2FC$ | $padj$   | $baseMean$ | Description                          |
|-----------|------|------------|----------|------------|--------------------------------------|
| HRNR      | ↑    | +3.12      | 5.48e-17 | 70.61      | Hornerin                             |
| FLG       | ↑    | +2.90      | 9.26e-08 | 30.17      | Filaggrin                            |
| GJA1      | ↑    | +2.51      | 2.78e-03 | 74.30      | Gap junction protein alpha 1         |
| CEND1     | ↑    | +2.49      | 3.65e-02 | 55.11      | Cell cycle exit and neuronal diff. 1 |
| SCN2A     | ↑    | +2.14      | 1.22e-04 | 39.66      | Sodium voltage-gated channel alpha 2 |
| HBA2      | ↓    | -2.10      | 9.50e-11 | 25.00      | Hemoglobin subunit alpha 2           |
| HBA1      | ↓    | -1.82      | 1.40e-09 | 29.52      | Hemoglobin subunit alpha 1           |
| LINC00887 | ↓    | -1.80      | 4.17e-04 | 25.95      | Long intergenic non-coding RNA 887   |
| LCNL1     | ↓    | -1.78      | 2.87e-05 | 15.97      | Lipocalin like 1                     |
| NDNF      | ↓    | -1.78      | 3.20e-04 | 16.63      | Neuron derived neurotrophic factor   |

Table 3. Top 5 upregulated (↑) and top 5 downregulated (↓) DEGs.  $\log_2FC$  =  $\log_2$  fold change;  $padj$  = BH-adjusted p-value;  $baseMean$  = mean normalized counts.

### 3.6 KEGG Pathway Enrichment

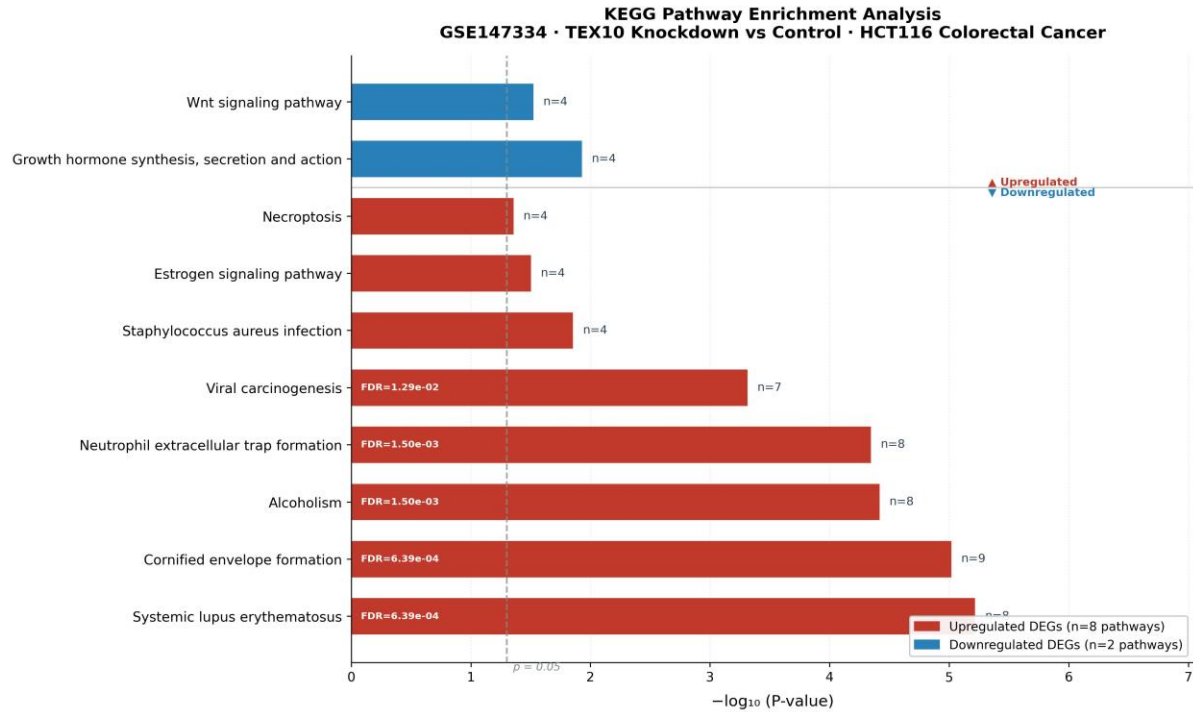
Upregulated DEG analysis via DAVID identified 8 KEGG pathways at  $EASE \leq 0.05$  (Table 4). Five achieved Benjamini  $FDR < 0.05$ : Systemic lupus erythematosus (8 genes,  $FDR = 6.39 \times 10^{-4}$ , fold enrichment = 10.99), Cornified envelope formation (9 genes,  $FDR = 6.39 \times 10^{-4}$ ), Alcoholism (8 genes,  $FDR = 1.50 \times 10^{-3}$ ), Neutrophil extracellular trap formation (8 genes,  $FDR = 1.50 \times 10^{-3}$ ), and Viral carcinogenesis (7 genes,  $FDR = 1.29 \times 10^{-2}$ ). Downregulated DEG analysis identified 2 nominally significant pathways: Growth hormone synthesis and action (4 genes,  $p = 0.0117$ ) and Wnt signaling pathway (4 genes,  $p = 0.0298$ ).

| KEGG Pathway                            | Dir. | Count | P-Value  | Benjamini FDR      | Fold Enrichment |
|-----------------------------------------|------|-------|----------|--------------------|-----------------|
| Systemic lupus erythematosus            | ↑    | 8     | 6.03e-06 | <b>6.39e-04 **</b> | 10.99           |
| Cornified envelope formation            | ↑    | 9     | 9.54e-06 | <b>6.39e-04 **</b> | 8.21            |
| Alcoholism                              | ↑    | 8     | 3.80e-05 | <b>1.50e-03 **</b> | 8.29            |
| Neutrophil extracellular trap formation | ↑    | 8     | 4.48e-05 | <b>1.50e-03 **</b> | 8.07            |
| Viral carcinogenesis                    | ↑    | 7     | 4.83e-04 | <b>1.29e-02 *</b>  | 6.76            |
| Staphylococcus aureus infection         | ↑    | 4     | 1.39e-02 | 3.10e-01           | 7.76            |
| Estrogen signaling pathway              | ↑    | 4     | 3.12e-02 | 5.96e-01           | 5.69            |
| Necroptosis                             | ↑    | 4     | 4.37e-02 | 7.32e-01           | 4.98            |
| Growth hormone synthesis & action       | ↓    | 4     | 1.17e-02 | 1.00e+00           | 8.19            |
| Wnt signaling pathway                   | ↓    | 4     | 2.98e-02 | 1.00e+00           | 5.74            |

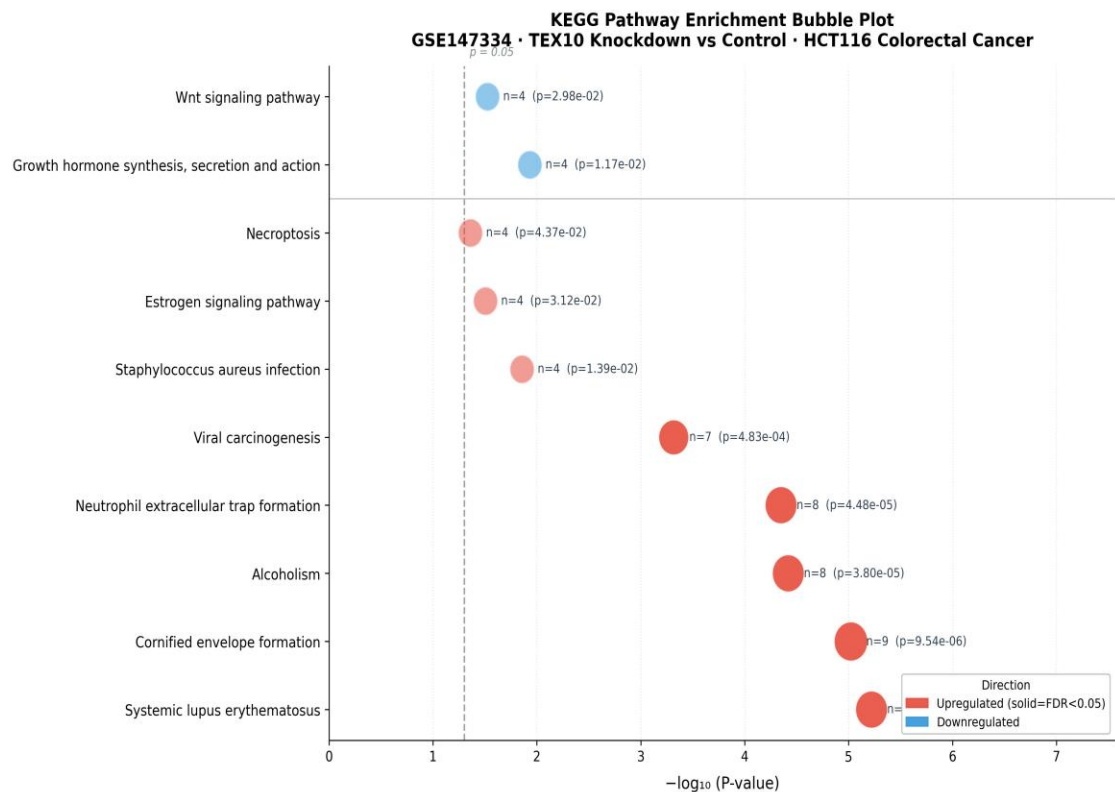
Table 4. KEGG pathway enrichment results from DAVID analysis. ↑ = upregulated; ↓ = downregulated DEGs. \*  $FDR < 0.05$ ; \*\*  $FDR < 0.01$ . Count = number of DEGs per pathway.

### 3.7 KEGG Pathway Visualizations

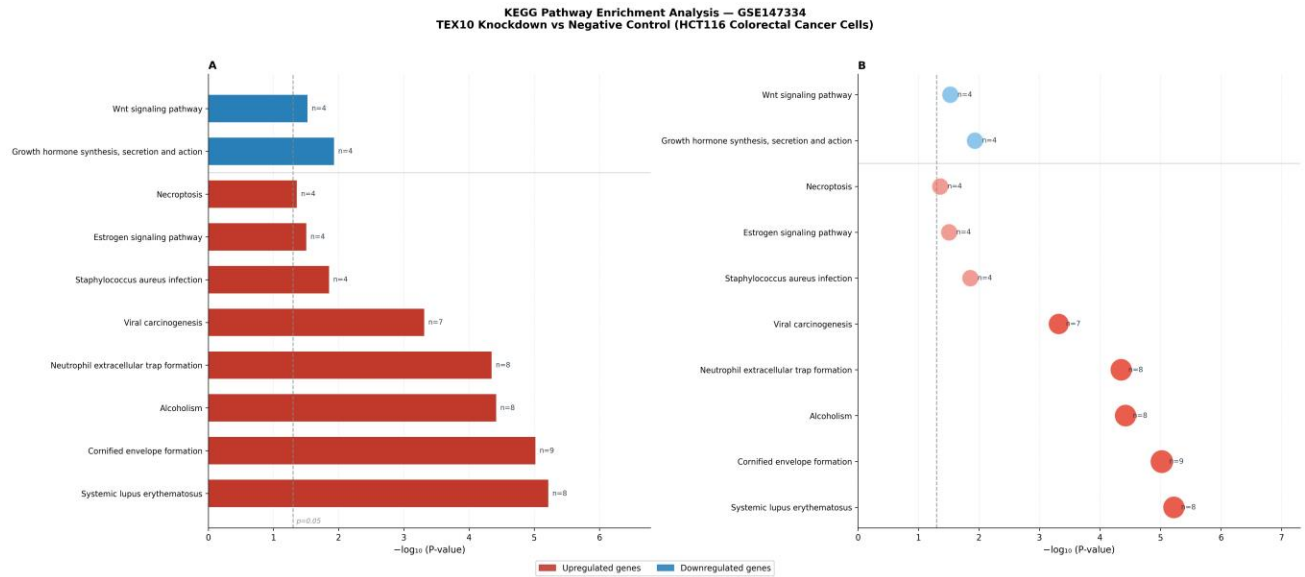
Figures 8–10 present KEGG enrichment results. The bar chart (Figure 8) ranks pathways by  $-\log_{10}(p\text{-value})$  with FDR values labeled inside bars for Benjamini  $\leq 0.05$ . The bubble plot (Figure 9) adds gene count information via bubble size. The combined figure (Figure 10, Panel A = bar chart, Panel B = bubble plot) presents both in publication-ready layout.



**Figure 8.** KEGG pathway enrichment bar chart (GSE147334, TEX10 Knockdown vs. Control, HCT116 CRC). X-axis:  $-\log_{10}(P\text{-value})$ . Red bars = upregulated DEG pathways (n=8); Blue bars = downregulated DEG pathways (n=2). Gene counts labeled as n=. Benjamini FDR annotated where  $\leq 0.05$ . Dashed line:  $p = 0.05$ .



**Figure 9.** KEGG pathway enrichment bubble plot (GSE147334). X-axis:  $-\log_{10}(P\text{-value})$ ; Y-axis: KEGG pathways. Bubble size = number of DEGs per pathway. Red = upregulated (solid = FDR < 0.05); Blue = downregulated. P-values annotated per pathway.



**Figure 10.** Combined KEGG enrichment figure (GSE147334). Panel A = bar chart; Panel B = bubble plot. Red = upregulated gene sets; Blue = downregulated gene sets. Both panels share the same pathway ranking by  $-\log_{10}(P\text{-value})$ .

## 4. DISCUSSION

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This study presents a systematic GEO2R-based transcriptomic analysis of TEX10-depleted HCT116 colorectal cancer cells using dataset GSE147334. Analysis of 18,903 genes identified 235 significant DEGs (103 upregulated, 132 downregulated) and 10 enriched KEGG pathways, providing a comprehensive molecular map of the transcriptomic consequences of TEX10 loss.

The most highly upregulated gene, HRNR (hornerin;  $\log_2\text{FC} = +3.12$ ), encodes a member of the S100 fused-type protein family involved in epidermal cornification. Its upregulation is consistent with Cornified envelope formation ( $\text{FDR} = 6.39 \times 10^{-4}$ ) as a top enriched pathway, suggesting TEX10 suppresses an aberrant cornification gene program in CRC. FLG (filaggrin;  $\log_2\text{FC} = +2.90$ ), co-enriched in this pathway, is a structural barrier protein whose upregulation upon TEX10 knockdown may alter cell adhesion properties relevant to tumor invasiveness.

GJA1 (connexin 43;  $\log_2\text{FC} = +2.51$ ) encodes a gap junction protein with established tumor suppressor roles in CRC, where gap junction-mediated intercellular communication restrains tumor cell proliferation. Its upregulation upon TEX10 depletion supports reduced proliferative capacity consistent with the experimental observations.

Among downregulated genes, HBA2 and HBA1 ( $\log_2\text{FC} = -2.10$  and  $-1.82$ ) encode hemoglobin subunits whose expression in colorectal cancer cells has been linked to oxygen sensing and redox homeostasis. Their suppression may reflect metabolic reprogramming following TEX10 depletion. The significant downregulation of EGR1 ( $\log_2\text{FC} = -1.71$ ,  $\text{padj} = 1.14 \times 10^{-12}$ ,  $\text{baseMean} = 5,763$ ) — a zinc finger transcription factor regulating NF- $\kappa$ B-related gene networks — suggests TEX10 normally maintains EGR1 expression as part of the NF- $\kappa$ B regulatory axis.

Viral carcinogenesis enrichment (7 genes,  $\text{FDR} = 1.29 \times 10^{-2}$ ) is consistent with NF- $\kappa$ B dysregulation in cancer. Neutrophil extracellular trap (NET) formation enrichment (8 genes,  $\text{FDR} = 1.50 \times 10^{-3}$ ) suggests TEX10 modulates innate immune gene programs in CRC cells, an emerging area with immunotherapeutic implications.

Wnt signaling pathway enrichment among downregulated genes ( $p = 0.0298$ ), though nominally significant, is biologically important — Wnt/ $\beta$ -catenin is the most frequently activated pathway in CRC with APC mutations in >80% of sporadic cases. Downregulation of Wnt

components upon TEX10 knockdown suggests TEX10-NF- $\kappa$ B/Wnt crosstalk deserving experimental characterization.

Seven quality control visualizations confirmed analytical rigor: the box plot validated normalization uniformity, the p-value histogram confirmed genuine differential signal, the dispersion plot validated DESeq2 model fit, and the UMAP confirmed clear transcriptomic group separation. Limitations include the cell-line basis and the knockdown (rather than tumor-vs-normal tissue) experimental design, which limits direct clinical translatability. Future studies should validate key findings in patient-derived organoids and in vivo models.

## 5. CONCLUSION

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In this comprehensive bioinformatics study, GEO2R/DESeq2 analysis of GSE147334 identified 235 significant DEGs (103 upregulated, 132 downregulated) from 18,903 tested genes in TEX10 knockdown versus shNC control HCT116 colorectal cancer cells. Seven quality control figures confirmed data integrity and group separation. The top upregulated gene HRNR ( $\log_2FC = +3.12$ ) and FDR-significant enrichment of Cornified envelope formation ( $FDR = 6.39 \times 10^{-4}$ ) and Viral carcinogenesis ( $FDR = 1.29 \times 10^{-2}$ ) implicate TEX10 in suppressing aberrant epithelial differentiation and modulating immune gene programs in CRC. Downregulation of EGR1 and nominal Wnt signaling enrichment indicate broad transcriptional reprogramming beyond canonical NF- $\kappa$ B. Three publication-quality KEGG visualizations (bar chart, bubble plot, combined figure) effectively communicate pathway-level findings. Collectively, these results establish a transcriptomic framework for understanding TEX10 function in CRC and identify GJA1, EGR1, and Wnt pathway components as priority candidates for experimental validation.

## 6. REFERENCES

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